Milestone 3: LLM Experimentation and Model Selection

Larissa Decker & Serena Zhou

1. LLM Integration Points in System Architecture

Our system uses LLMs at two specific points in the processing pipeline, each with distinct requirements:

1. Hypothesis Generation: Structured output with no hallucinations.
2. Business Summary: Clear actionable recommendations.

2. Models Evaluated:

Model	Type	Cost (per 1M tokens)	Access Method
Claude 3.5 Sonnet	Commercial	$3 input / $15 output	Anthropic API
GPT-5	Commercial	$2.50 input / $10 output	OpenAI API
Local Ollama	Open Source	Free	Local

3. Test Scenario:

We tested each model on the same representative prompts grounded in our use case.
Each prompt included

- a detected KPI anomaly
- summary statistics from automated analysis
- constraints on hallucinations

Base Prompt (example to adapt)

You are a KPI root-cause analysis assistant. Use ONLY the facts provided.
If data is insufficient, say so and propose checks.

Input: KPI anomaly summary and statistical findings already computed

Instructions

1. Base all claims strictly on the provided data
2. clearly separate evidence from hypotheses
3. write for a senior executive audience
4. do not introduce causes not supported by the data

Output format:

- executive summary (2-3) sentences
- key evidence

- plausible explanations (clearly labeled as hypotheses)
- suggested next analysis

Dataset Context: E-commerce sales data with the following columns:
date, revenue, orders, traffic, conversion_rate, avg_order_value, region, device_type, marketing_spend, product_category
Detected Anomaly: Revenue dropped 15% starting April 20th, 2024 ($p < 0.01$)

3a. Task 1 Hypothesis Generation Prompt
You are a data analysis assistant. Given a dataset and a detected KPI anomaly, generate hypotheses for what might have caused the change. CRITICAL: Only reference columns that exist in the dataset. Do not invent metrics.

Dataset columns: date, revenue, orders, traffic, conversion_rate, avg_order_value, region, device_type, marketing_spend, product_category

Anomaly: Revenue dropped 15% starting April 20, 2024

Generate 3 to 5 testable hypotheses. For each, specify which column(s) to analyze.

3b. Task 2 Business Summary Prompt
You are a business analyst. Summarize these statistical findings for an executive audience in 3 to 4 sentences. Be clear about confidence levels and suggest next steps.

Findings: Primary driver: Mobile conversion_rate dropped 22% ($r=0.71$, $p<0.01$). Secondary: marketing_spend in the West region decreased 35%. Ruled out: avg_order_value (no significant change), product_category mix (stable).

4. Model Output Comparison
Claude Output:
Task 1: Hypothesis Generation Outputs

- Hypothesis 1: Traffic Decline – analyze traffic, date
- Hypothesis 2: Conversion Rate Drop – analyze conversion_rate, device_type, date
- Hypothesis 3: Lower Average Order Value – analyze avg_order_value, product_category, date
- Hypothesis 4: Regional Performance Shift – analyze revenue, region, date
- Hypothesis 5: Marketing Spend Reduction – analyze marketing_spend, traffic, region, date

Task 2: Business Summary Outputs
Claude:
The 15% revenue decline is primarily driven by a 22% drop in mobile conversion rate ($r=0.71$, $p<0.01$). A secondary factor is the 35% reduction in West region marketing spend. Avg_order_value and product_category mix were ruled out. Next steps: Investigate mobile checkout experience and evaluate restoring West region marketing spend.

GPT Output
Executive Summary
Revenue declined by 15% starting April 20, 2024, driven primarily by a sharp deterioration in mobile conversion rate and compounded by a reduction in marketing spend in the West region. Other core revenue components—average order value and product mix—remained stable and do not explain the decline.

Key Evidence
- Mobile conversion rate dropped 22%, showing a strong association with revenue changes ($r = 0.71$, $p < 0.01$), indicating this is the primary statistical driver.
- Marketing spend in the West region decreased 35% over the same period, aligning temporally with the revenue decline and acting as a secondary contributor.
- Average order value showed no statistically meaningful change.
- Product category mix remained stable, ruling out shifts in assortment as a factor.

Plausible Explanations (Hypotheses)
- The decline in mobile conversion rate may reflect a disruption or degradation in the mobile purchasing funnel beginning around April 20, 2024.
- The reduction in West-region marketing spend may have lowered traffic quality or volume in that region, amplifying the revenue impact of lower conversion.

(These are hypotheses consistent with observed patterns; causal mechanisms are not directly identified in the data.)

Suggested Next Analysis
- Decompose the mobile conversion rate decline by region and product category to assess whether the issue is localized or systemic.
- Examine traffic volume and traffic source trends for mobile users before and after April 20 to determine whether the conversion drop reflects demand quality versus on-site performance.
- Conduct a pre/post analysis of West-region revenue and traffic elasticity relative to marketing spend to quantify incremental impact.

Local Ollama Output
% time ollama run qwen2.5:7b-instruct < ~/LLM-3.txt
Executive Summary:
Revenue declined by 15% starting from April 20, 2024, primarily due to a significant drop in the mobile conversion rate. The
The West region's reduced marketing spend also contributed, but was not the primary driver.
Key Evidence:
- Mobile conversion_rate dropped 22%, with a strong correlation (r=0.71) and statistical significance (p<0.01).
- Marketing_spend in the West region decreased by 35%.
- No significant change in avg_order_value or product_category mix.
Plausible Explanations:
Hypothesis 1: The drop in mobile conversion_rate may be attributed to recent changes in mobile user experience or design
issues on the mobile app/website, as these changes typically affect this metric significantly.
Hypothesis 2: A shift in customer behavior towards desktop usage could have occurred, though desktop data is not available
here. This would support the drop in mobile conversion_rate while potentially explaining why avg_order_value remained stable.
Suggested Next Analysis:
- Conduct a detailed user experience (UX) review of the mobile app/website to identify and address any issues.
- Collect and analyze desktop traffic and conversion rates to determine if there has been a shift in customer behavior.
ollama run qwen2.5:7b-instruct < ~/LLM-3.txt 0.06s user 0.16s system 0% cpu 37.688 total

6. Model Selection Rationale

Selected Model: Based on the following tradeoffs, GPT was selected as the primary model for the prototype use case

1. Output Quality: GPT showed the highest fidelity to provided data; Claude performed well but was slightly less strict about honoring ruled-out variables; Ollama also exhibited speculative drift.

2. Cost: API-based models (GPT, Claude) incur per-call costs but require minimal engineering and oversight; local models have low marginal inference cost but higher hidden costs related to prompt engineering and monitoring.
3. Latency/Performance: GPT and Claude produced responses within seconds; Local Ollama model exhibited higher latency.

7. Baseline Performance Metrics

The following baseline metrics will be used to evaluate system performance throughout development:

Quantitative Baseline:

- evidence fidelity: 100% of claims directly supported by provided findings
- hallucination rate: near zero unsupported causal claims
- output structure: 100% adherence to required response format
- latency: low (seconds per response)

Qualitative Baseline:

- executive readability: high, i.e., clear summaries, concise framing, no jargons
- casual claims: clear distinction between evidence and hypotheses
- actionability: consistent generation of analytically grounded next steps